INSTRUCTOR Solution

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

Time = 15 Minutes **Quiz No 7** [20 points]

Note - *Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.*

Question 1 [10 points] Independence & Conditional Independence

The joint distribution for the two DRVs, X & Y, is given below.

Y = Second roll \ X = First roll	1	2	3	4
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		

(a) **[3 points]** Are the two R.Vs X and Y independent?
Note - You need to clearly highlight the conditions for independence and justify your answer mathematically. ***The points are not for the right answer but for the working around the justification of the right answer.***

(b) **[7 points]** Are the two R.Vs X and Y independent, conditioned on event A, where event A = $X \leq 2$ and $Y \geq 3$?
Note - You need to clearly highlight the conditions for the independence and justify your answer mathematically. ***The points are not for the right answer but for the working around the justification of the right answer.***

Question 2 [10 points] Calculating Probability for Discrete Random Variables

A Random Variable Z_n, is defined as $Z_n = \min(X_1, X_2, \ldots X_n)$, where X_i's are outcomes of identical and independently distributed fair 6-sided die. As an example, $Z_2 = \min(X_1, X_2)$ – meaning Z_2 is the minimum of the outcome of the two independent throws of the 6-sided die X_1, X_2.

(a) **[5 points]** Calculate the probability that $P(1 < Z_{12} < 6)$.

(b) **[5 points]** Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$.

1 (a) for RV X, Y to be independent $P_{XY}(X,Y) = P_X(x) \cdot P_Y(y)$ for all x, y.

3 pts

or $P_X(x) = P_{X|Y}(x|y)$ for all x, y 1

we see that $P_X(1) = \frac{3}{20}$

while $P_{X|Y}(1|Y=1) = 0$ 2 pts.

Therefore, X and Y are NOT Independent.

1(b) $A = \{X \leq 2 \text{ and } Y \geq 3\}$

$P_{X,Y|A}(x,y|A)$

Y = Second Roll		
4	1/9	2/9
3	2/9	4/9
	1	2

X = first Roll

2 pts

2 pts

$$P_{X|A}(1|A) = 1/3, \quad P_{Y|A}(3|A) = 2/3$$

$$P_{X|A}(2|A) = 2/3, \quad P_{Y|A}(4|A) = 1/3$$

i) $P_{X,Y|A}(1,3|A) = \frac{2}{9} = P_{X|A}(1|A) \cdot P_{Y|A}(3|A)$

(A) (B)

$= 1/3 \cdot 2/3$

$= 2/9$ LHS = RHS

3pts

ii) $P_{X,Y|A}(1,4|A) = 1/9$

& $P_{X|A}(1|A) \cdot P_{Y|A}(4|A) = 1/3 \cdot 1/3 = 1/9 \Rightarrow$ LHS = RHS

iii) $P_{X,Y|A}(2,3) = \frac{4}{9}$ and $P_{X|A}(2|A) \cdot P_{Y|A}(3|A)$

$= 2/3 \cdot 2/3 = 4/9$ LHS = RHS

(iv) $P_{X,Y|A}(2,4) = 2/9$ and $P_{X|A}(2|A) \cdot P_{Y|A}(4|A)$

$= \frac{2}{3} \cdot 1/3 = \frac{2}{9}$

LHS . RHS .

So $P_{X,Y|A}(x,y|A)$ where $A = \{x \leq 2 \text{ and } y \geq 3\}$

are Conditionally Independent.

2 (a) $P(1 < Z_{12} < 6)$

5pts

$= 1 - (P_{Z_{12}} = 1) - P(Z_{12} = 6)$ —— 1pt

$P(Z_{12} = 1) = 1 - P(Z_{12} \neq 1)$

$= 1 - P(Z_{12} \geqslant 2)$; all 12 throws should be 2,3,4,5 or 6

$P(Z_{12} = 1) \quad = 1 - \left(\frac{5}{6}\right)^{12}$ —— 1pt

$P(Z_{12} = 6) = P(\text{all throws have } 6)$

$= \left(\frac{1}{6}\right)^{12}$ —— 1pt

$P(1 < Z_{12} < 6) = 1 - \left(1 - \left(\frac{5}{6}\right)^{12}\right) - \left(\frac{1}{6}\right)^{12}$

$= \left(\frac{5}{6}\right)^{12} - \left(\frac{1}{6}\right)^{12}$ —— 1pt

$= (0.833)^{12} - (0.1666)^{12}$

$= 0.11215 - 0.0000000000459$ —— 1pt

$= 0.1121566$

(b) $P(Z_n = 1) = 1 - P(Z_n \neq 1)$ —— 1pt

5pts

$= 1 - P(\text{all } n \text{ throws are greater than } 1)$

$= 1 - (5/6)^n$ —— 1pt

$\Rightarrow 1 - \left(\frac{5}{6}\right)^n \geqslant 0.6$ —— 1pt

$(5/6)^n \leq 0.4 \Rightarrow n \log 5/6 \leq \log 0.4$

$n \geqslant \frac{-0.397}{-0.079}$

$n \geqslant 5.02 \implies n = 6 \text{ (min)}.$ 2pt